

Tiruchirappalli City Transit Portal

Comprehensive Deployment, User Manual & Architectural Reference

1. Project Executive Overview

The **Tiruchirappalli City Transit Portal** is an enterprise day pass management and real-time fleet tracking application tailored specifically for the public bus network of Tiruchirappalli (Trichy), Tamil Nadu, India. It integrates cryptographic QR-code ticketing with a dual-mode telemetry engine: an innovative **device-clock transit simulation** matching authentic bus schedules, and a **smartphone GPS telemetry cockpit** streaming genuine satellite telemetry via HTML5 geolocation with automatic 30-second disconnect fallbacks.

2. Prerequisites & System Requirements

Component	Minimum Requirement	Recommended
Operating System	Windows 10/11, macOS 11+, Linux	Any modern 64-bit OS
Python Version	Python 3.9+	Python 3.11 / 3.12 / 3.14
Web Browser	Chrome 90+, Edge 90+, Firefox, Safari	Latest Chrome or Edge with GPS enabled
Git	Git 2.25+	Latest Git CLI
Network Port	TCP Port 5000 free	Localhost / Wi-Fi LAN

3. How to Start the Project (Quickstart Guide)

Step 1: Clone or Open the Repository:

```
git clone https://github.com/gsk37690-tech/Bus-Pass-and-Live-Tracking-Software.git
cd Bus-Pass-and-Live-Tracking-Software
```

Step 2: Install Python Dependencies:

```
pip install -r requirements.txt
(Packages: Flask, qrcode, Pillow, gunicorn, reportlab)
```

Step 3: Run the Web Server:

```
python web_app.py
```

The server will start on **http://127.0.0.1:5000**.

Step 4: Run Automated System Tests (Optional Self-Check):

```
python -m unittest tests/test_system.py
```

Executes all 21 automated integration tests (Auth, Routing, Telemetry, Ticketing, Deletion).

4. Core Features & Usage Instructions

A. Dual-Portal Authentication (/login)

Separates Passengers and Bus Drivers with session management. Features 1-click quick logins:

- **Passenger:** P001234 / pass123 (John Doe - Active Pass)
- **Driver:** DRV-101 / drv123 (R. Murugan - Central Depot)

Directs commuters to the Commuter Dashboard and drivers to the mobile Cockpit.

B. Tiruchirappalli (Trichy) Route Network

Localized with authentic GPS coordinates and stops:

- **Route 101:** Central Bus Stand (CBS) ■ Chathiram Bus Stand (Palakkarai, Thillai Nagar, Main Guard Gate)
- **Route 204:** CBS ■ Srirangam Rajagopuram (HPO, Anna Statue, Cauvery Bridge, Thiruvanaikoil)
- **Route 305:** CBS ■ NIT Trichy / BHEL (TVS Tollgate, Golden Rock, Thiruverumbur)
- **Route 402:** Chathiram ■ Tiruchirappalli International Airport TRZ (Mannarpuram)

C. Device-Clock Transit Simulation Engine

Replaces artificial sleep-interval loops with mathematical progression. Reads device local time (HH:MM:SS) and calculates the exact interpolated coordinates (lat, lng), remaining distance, and ETA between stops along the schedule.

D. Mobile Driver Cockpit & Smartphone Satellite GPS (/driver)

Mobile-optimized dashboard mount interface:

- **Start/Stop Trip toggle:** Streams HTML5 geolocation (watchPosition) high-accuracy coordinates.
- **HUD Metrics:** Live Speed (km/h), GPS Precision ($\pm$ meters), Ping counter, Geodesic coords.
- **Haversine Stop Detection:** Automatically identifies the nearest station and distance in meters.
- **Screen Wake-Lock API:** Prevents mobile screen sleep during driving.
- **30-Second Fallback:** Gracefully transitions back to clock schedule if driver stream stops.
- **Onboard Ticket Inspector:** Drivers can scan/enter pass tokens at boarding doors.

E. Commuter Live Fleet Map & Stop Progression Timeline (/)

Interactive Leaflet map centered at Trichy (10.7956, 78.6856). Displays dynamic status badges:

- ■ SATELLITE GPS BROADCAST (Phone Active) when a driver is streaming phone GPS.
- ■ REAL-TIME CLOCK SYNC (Trichy Timetable) when in schedule mode.

F. Digital Day Pass Management & QR Wallet

Issue Day Passes, 24-Hour Tourist Passes, Student and Weekend passes. Generates cryptographically verifiable QR codes with PNG rendering and Base64 embedding.

G. QR Ticket Inspector (/ & Ticket Tab)

Validates QR tokens against pass status, issue date, and expiration timestamp. Records validation audit events.

H. Audit Ledger & Pass Expiry Management

Full transaction and boarding ledger with live validity checks (Active, Expiring Soon, Expired). Features dedicated single-click  **Delete Row** inside the Check Pass popup modal next to Renew Pass.

5. System Verification & Key URLs

Interface / Endpoint	URL	Function
Commuter Dashboard	http://127.0.0.1:5000/	Interactive Map, Passes, Ledger
Login Portal	http://127.0.0.1:5000/login	Role auth (Passenger / Driver)
Driver Cockpit	http://127.0.0.1:5000/driver	Smartphone GPS broadcast & HUD
Live Telemetry API	http://127.0.0.1:5000/api/bus-live/BUS-101	JSON coordinates & timeline
Health Diagnostic	http://127.0.0.1:5000/api/health	Fleet counts & DB integrity